



An effective and unified method to derive the inverse kinematics formulas of general six-DOF manipulator with simple geometry

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ABSTRACT

For decades, although the inverse kinematics (IK) problem of six degrees of freedom (DOF) manipulator with general geometry has been solved, there is still needed an efficient and unified method to derive the IK formulas for the six-DOF manipulator with simple geometry. In this paper, we propose an improved method of disconnection and re-connection to achieve this purpose. Firstly, we determined the position selection of the two cutting points and the attitude setting at the first cutting point. Next, we gave the new conditions for the re-connection of the two sub-chains and the corresponding equations. Then, we summarized five methods of cutting the manipulator into two three-DOF sub-chains and corresponding equation deformations, as well as the selection principles of equation deformations. Finally, we used three examples to demonstrate the effectiveness of the method. The results show that the expressions of six joint variables can be derived from only six equations. The solutions of the wrist joint variables are not affected by the singularity.

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1. Introduction

Robots integrate the theoretical knowledge and technological achievements of many disciplines, such as mechanism [1–3], computer imaging science [4–6], intelligent control [7,8], pattern recognition [9–13], artificial intelligence [14–16], signal processing [17,18], etc., and has been a research hot-spot and direction target in the technical field since its birth.

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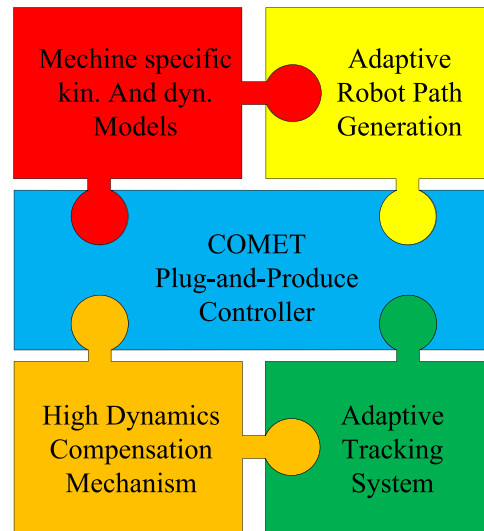


Fig. 1. Precision machining scheme for industrial robots.

Although with the development of sensing technology, especially machine vision [19], robots are gradually extending to electronics [20], chemical industry [21], medical and health care [22–24], agriculture [25,26], and other areas, robots for industrial use still have a large market share. The EU/FP7 project COMET [27,28] proposed a robot application in the high-end processing field, as shown in Fig. 1. This research includes the study of kinematic and dynamic methods to describe industrial robots and accurately define an industrial robot's static and dynamic behavior. The second is the integrated programming and simulation environment for the path generation of the adaptive robot. Besides, an adaptive tracking system for industrial robots is proposed to detect the robot's execution path's deviation. Finally, the high dynamic compensation mechanism is studied to achieve the absolute accuracy of more than $50\ \mu\text{m}$. To realize this project, solving the inverse kinematics (IK) problem of the six degrees of freedom (DOF) serial manipulator is essential, which is the premise of motion control, trajectory planning, and obstacle avoidance. In his Ph.D. thesis, Pieper proposed a design criterion for a class of 6R serial manipulators with analytical solutions. The axes of three consecutive joints intersect at the same point or parallel each other [29]. Since the analytical solutions are the most efficient in the IK methods, the 6R spherical wrist industrial robot, which has been widely applied in industry, is designed according to the Pieper criterion in its last three joints.

However, with the increasing demand for robots in human production, another kind of six-DOF serial manipulator, which does not meet the Pieper criterion, has been gradually applied to spraying, welding, and material handling [30]. It is collectively referred to as the general six-DOF manipulator that the adjacent joint axes' spatial position relationship and the connecting links' length are arbitrary. The coupling degree and complexity of its kinematics equations are higher than that of the sphere-wrist structure. Its IK problem was called "Mount Everest of Kinematics Problem" by Freudenstein [31] and has attracted many excellent scholars who devoted themselves to solving it. In 1986, Liao et al. [32] first obtained its sixteen IK solutions using the unitary matrix method. A year later, the same result was obtained by LEE [33] using vector theory. Following their work, Raghavan and Roth [34] derived a unitary polynomial of degree 16, with its root being the IK solution of a joint using dialysis elimination. Since then, many researchers have worked on improving the algorithm [35–37]. In 2007, Husty et al. [38] obtained the univariate 16th-degree polynomial solution by combining the theory of multidimensional space geometry with the classical Segre clustering theory. Qiao et al. [39] then solved the problem using double quaternions and Dixon resultant for the first time. While these methods can solve all the inverse kinematics problems of the non-redundant six-DOF manipulator, they can hardly be implemented in the industrial manipulator due to their highly complicated and time-consuming derivation procedure [40–42]. In practical, the general six-DOF manipulators are usually designed with simple geometric structures. They have at least one pair of adjacent axes intersecting at a point; except for that, the other adjacent axes are only perpendicular or parallel to each other. Despite their simple structure, a universal and efficient solution to their IK problem is needed. A practical approach should include the following aspects. First, the transformation of the forward kinematics equation (FKE) can be efficiently determined. Next, the closure equations to be derived can be precisely found, rather than relying on intuition or luck, and they contain the fewest number of variables. Then, each joint variable's derivation process is intuitive and convenient, and the expression is straightforward. Finally, the algorithm can meet the requirements of high precision and real-time. In this paper, we propose an improved double cutoff points method to achieve these goals.

This paper's remainder is organized as follows: Section II summarizes the work done by other researchers related to this method. Section III introduces selecting the cut points and the corresponding forward kinematic equation deformation, and

the IK formula derivation steps. Section IV presents the implementation of the proposed IK method on three manipulators. Finally, some conclusions and prospects are drawn in Section V.

2. Related works

It exists a long time that the concept of the serial-chain manipulator breaking into two sub-chains. In 1984, Lee and Ziegler [43] proposed to separate the arm and wrist of the PUMA robot and then calculate the three joint variables in the positioning structure with geometric methods. On this basis, the solutions of the three wrist joint variables are obtained. Subsequently, Brandstötter et al. [44] use this method to get the analytical solutions of spherical wrist industrial 6R manipulator. However, their disconnection method is not optimal, and the calculations of the fourth and sixth joints would not be possible due to wrist singularity. Trinh et al. [30] solved the IK problem of two non-spherical wrist manipulators in the same decomposition way. They considered the geometric relationship between the adjacent axes at the point of decomposition. Unfortunately, their approach does not intuitively determine closure equations, which need to be found through complex geometric relationships. Based on the algorithm of [38], Pfurner [45] proposed to cut the 6R-chain into a 4R-chain and a 2R-chain to solve the IK problem of the special manipulators. In his example, the manipulator structure is now widely applied on the universal robot (UR) [46], which has three consecutive axes parallel to each other. Both [45] and [46] derive the first joint variable and then use it to calculate other joint variables. Still, the derivation process and the resulting expression in [45] are far more complicated than those in [46]. That implies that choosing the right cut point can simplify the derivation and the IK expression. The above literature pieces only use one cut point, which cannot make full use of the manipulator's geometric information. Lee and Reinholtz [47] indicate that selecting two appropriate joints' origin as cut points can make the closure equations have the fewest variables and make the equation more concise. However, they only pointed out this phenomenon, which does not develop a method to choose the cut point. In pieces of literature [42,48,49], a particular joint variable is selected to express others, and then a unary nonlinear equation about this joint variable is obtained. The nonlinear equation solutions are obtained by one-dimensional search, and the other joint angles are obtained by inverse substitution. It is suitable for use in controllers of the commercial manipulator. The FKE transformation they adopted is derived from literature [34], which can be seen as cutting off at the third and fifth joints' origin and decomposing the manipulator into two three-DOF sub-chains. Under this transformation, they lack general principles to guide the selection of the closure equations. If the transformation is applied directly to the manipulator in literature [41], it is challenging to realize the closure equations with the fewest variables. Based on [48,49], Bu et al. [50] try to find an appropriate cut point to simplify the formula derivation process and propose that the cut point be selected at the joint origin or intersection adjacent joint axes.

The deformation of FKE is equivalent to the decomposition of the manipulator into two sub-chains. Therefore, finding two suitable cutting points to decompose the manipulator is equivalent to finding the corresponding FKE deformation. Moreover, the geometric relationship between the adjacent joints' axes at the cut point is also essential and should be considered. Based on previous works, this paper develops a general method for selecting cut points and the attitude setting at the end of sub-chains to guide the FKE deformation based on the standard D-H [51] method. In the meantime, the new conditions for the re-connection of the sub-chains and their corresponding conditional equations are given. The plan is applied to three six-DOF manipulators with the structure proposed in this paper.

3. Solution process

3.1. Kinematics

The homogeneous transformation matrix of each joint should be defined first to obtain the six-DOF manipulator's kinematics. The standard D-H parameter method is used in this paper. The homogeneous transformation matrix of a single joint can be expressed as:

$$\begin{aligned} \mathbf{T}_i &= \text{Rot}(\theta_i, z_{i-1})\text{Trans}(0, 0, d_i)\text{Trans}(a_i, 0, 0)\text{Rot}(\alpha_i, x_i) \\ &= \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (1)$$

where

θ_i is the angle that rotates from x_{i-1} to x_i about z_{i-1} axis.

d_i is the distance move from x_{i-1} to x_i axes measured along z_{i-1} axis.

a_i is the distance move from z_{i-1} to z_i axes measured along x_i axis.

α_i is the angle that rotates from z_{i-1} to z_i about x_i axis.

The FKE of the end-effector (EE) relative to the base is obtained by multiplying all the homogeneous transformation matrices

$$\mathbf{T}_{\text{base_effector}} = \mathbf{T}_1(\theta_1)\mathbf{T}_2(\theta_2)\mathbf{T}_3(\theta_3)\mathbf{T}_4(\theta_4)\mathbf{T}_5(\theta_5)\mathbf{T}_6(\theta_6). \quad (2)$$

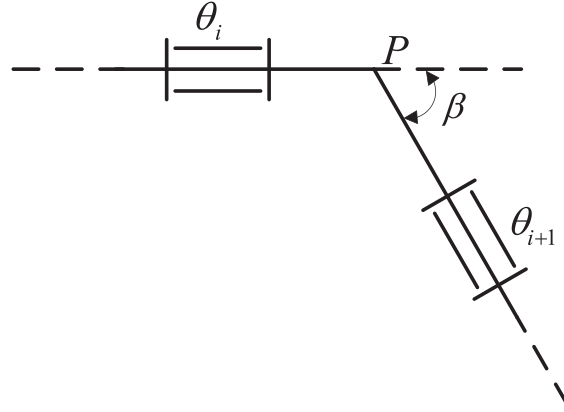


Fig. 2. Intersecting axes.

When given the value of joint variables in Eq. (2), an alternative representation of ${}^{base}_{end_effector} \mathbf{T}$ can be written as

$${}^{base}_{end_effector} \mathbf{T} = \begin{bmatrix} \vec{n} & \vec{o} & \vec{a} & \vec{p} & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (3)$$

Under the kinematics model established by the DH method, the inverse matrix transformation of Eq. (2) is usually carried out several times to achieve decoupling to obtain all the IK solutions corresponding to the given EE position and pose.

3.2. Disconnecting and reconnecting of the manipulator

Since a single homogeneous matrix can be left multiplied by the inverse transformation or right born by the inverse transformation, Eq. (2) has many kinds of deformations, not to mention the number of kinematic equations used to derive under each of these deformations. Fortunately, each of the transformation forms of Eq. (2) can be viewed as breaking the serial manipulator into two sub-chains and reconnecting them. Therefore, the optimal decomposition points obtained are equivalent to finding the equation's optimal transformation form (2). This section describes the serial manipulator's disconnection at two cut points and the attitude matrix setting at the first cut points.

3.2.1. Selection and analysis of the first cut point

This paper's method is mainly aimed at the general six-DOF manipulators with simple geometry: they have at least one pair of adjacent axes intersecting; except for that, the other adjacent joints' axis are only perpendicular or parallel to each other.

Literature [50] concluded that "the robotic kinematic chain is cut into two branches at the origin of the joint coordinate or at the intersection point of the axes of the targeted coordinates." From our perspective, it is best to choose the intersection point as the first cut point in intersecting axes.

Given the i th joint axis of a six-DOF manipulator intersects the $(i+1)$ th joint axis, $i=1,2,\dots,5$, shown in Fig. 2, P is the intersection point, scalar β is the angle between the adjacent axes. Take point P as the cut point to get two sub-chains and set it as the endpoint of the two sub-chains. For the sake of description, the one containing i th joint is denoted as the sub-chain L, and the other one having $(i+1)$ th joint is marked as the sub-chain R. As can be seen from the figure, the position of the endpoint P is not affected by the rotation of the two joints. The axial direction of the two joints is not affected by the rotation of the corresponding joint too. It is worth noting that if prismatic joints exist between intersecting axes, as long as they do not make the intersecting axes become different planes straight line in the process of moving, the above statement is still valid.

Remark 1. If the two sub-chains are separated by the intersection of the axes of the i th revolute joint and $(i+1)$ th joint, then the endpoint position equations of these two sub-chains and the equations of each axes are independent of θ_i and θ_{i+1} .

At this point, as long as the above conditions are satisfied, the four equations without θ_i and θ_{i+1} can be found, thus obtaining the IK solutions of the other four joints variable. Next, consider setting the attitude at the cut point so that one of the axes is collinear with the joint's corresponding axis.

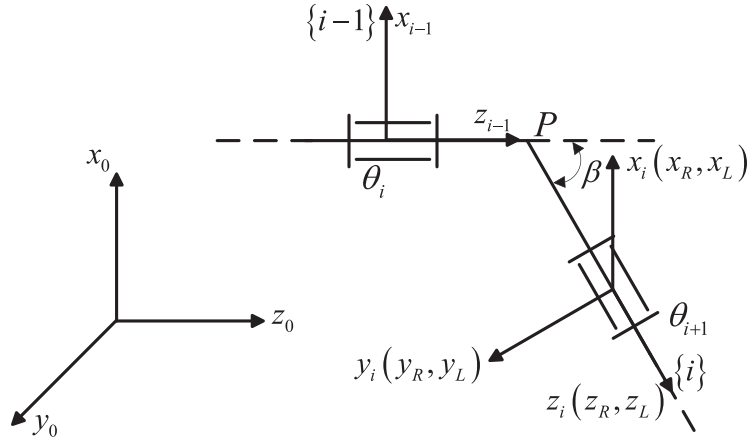


Fig. 3. The coordinate system of joint i .

3.2.2. Setting the attitude at the first cut point

The attitude setting at the first cut point needs to consider the value of the included angle β [52]. Since the standard D-H method is adopted in this paper to establish each joint's coordinate system, the i th joint's coordinate system is actually at the origin of the $(i+1)$ th joint, as shown in Fig. 3. When point P is taken as the separation point, the corresponding two sub-chains' terminal attitude in Eq. (2) will be consistent with the coordinate system $\{i\}$. Meanwhile, $\mathbf{z}_R$ of sub-chain R's terminal attitude axis is always collinear with the $(i+1)$ th joint axis. At this moment, the value of β affects the complexity of the sub-chain L's attitude equations.

Case 1: $\beta=0, \pi$. The axes of the two joints are collinear, and the terminal attitude of the two sub-chains has $\mathbf{z}_L=\mathbf{z}_R$. This design usually does not come up in practice, so it is not considered in this paper.

Case 2: $\beta=\pi/2$. $\mathbf{y}_L$ is collinear with the axis of the i th joint.

Case 3: $\beta \neq 0, \pm\pi/2$, and π . The terminal attitude axes of the sub-chain L are not collinear with any axes of the i th joint. When the manipulator is disconnected at point P , it is naturally proposed that the sub-chains' re-connection condition is that their end-pose coincide [45]. This condition can be used in **Case 2** to achieve our goal, but it is not effective in **Case 3** because there is no collinearity between the attitude axis of chain L and the i th joint axis. Therefore, to make $\mathbf{z}_L$ of sub-chain L in **Case 2** and **Case 3** collinear with the axis of the i th joint, according to Eq. (1), the terminal pose of the sub-chain L should be rotated about $\mathbf{x}_L$ (or $\mathbf{x}_i$) by $-\alpha_i$, that is, the terminal pose of the sub-chain L is $\mathbf{T}_L \bullet \text{Rot}(-\alpha_i, \mathbf{x}_i)$.

Mark: After the above operation, the unified condition for the two sub-chains' re-connection is: the position of the sub-chains' endpoints is coincident, and the included angle of their attitude remains unchanged compared with that before disconnection.

This condition is also uniformly expressed as equations (4)–(9)

$$\mathbf{p}_L = \mathbf{p}_R, \quad (4)$$

$$\mathbf{z}_L^T \mathbf{z}_R = \cos \beta, \quad (5)$$

$$\mathbf{y}_L^T \mathbf{z}_R = \cos \left(\frac{\pi}{2} - \beta \right) = \sin \beta, -180^\circ < \alpha < 0^\circ, \quad (6)$$

$$\mathbf{z}_L^T \mathbf{y}_R = \cos \left(\frac{\pi}{2} + \beta \right) = -\sin \beta, -180^\circ < \alpha < 0^\circ, \quad (7)$$

$$\mathbf{y}_L^T \mathbf{z}_R = \cos \left(\frac{\pi}{2} + \beta \right) = -\sin \beta, 0^\circ < \alpha < 180^\circ, \quad (8)$$

$$\mathbf{z}_L^T \mathbf{y}_R = \cos \left(\frac{\pi}{2} - \beta \right) = \sin \beta, 0^\circ < \alpha < 180^\circ, \quad (9)$$

Eqs. (4) and (5) are used to solve the four joint variables other than θ_i and θ_{i+1} . Eq. (6) or (8) is used to find θ_i , and Eq. (7) or (9) is used to find θ_{i+1} .

The selection of the first cut point and the attitude setting of the sub-chain L at this point can ensure that the equations used to solve the IK problem contain the fewest number of variables. However, to derive the final IK formula more efficiently, a second cut point needs to be selected.

3.2.3. Selection of the second cut point

After the first cut point is determined, the second cut point will be a natural thing. The six-DOF manipulators studied in this paper, except the intersection of adjacent rotating joints, other adjacent joints only exist parallel and perpendicular relation. Therefore, the second cut point is selected at the origin of a joint so that both sub-chains have three-DOF. The IK derivation method of the three-DOF manipulator [53,54] is relatively mature, and it can realize algebraic elimination quickly by using the existing results.

3.3. Selection of kinematic equation deformation

Based on the previous sections' results, Eq. (2) can be deformed in the following five ways regardless of how many sets of intersecting axes, the six-DOF manipulator has.

The axes of the first joint and the second joint intersect,

$$\begin{aligned}\mathbf{T}_L &= \mathbf{T}_5(\theta_5)\mathbf{T}_6(\theta_6)\mathbf{T}_1^{-1}(\theta_1)\text{Rot}(-\alpha_1, x_1) \\ \mathbf{T}_R &= \mathbf{T}_4^{-1}(\theta_4)\mathbf{T}_3^{-1}(\theta_3)\mathbf{T}_2^{-1}(\theta_2).\end{aligned}\quad (10)$$

The axes of the second joint and the third joint intersect,

$$\begin{aligned}\mathbf{T}_L &= \mathbf{T}_6(\theta_6)\mathbf{T}_1^{-1}(\theta_1)\mathbf{T}_2(\theta_2)\text{Rot}(-\alpha_2, x_2) \\ \mathbf{T}_R &= \mathbf{T}_5^{-1}(\theta_5)\mathbf{T}_4^{-1}(\theta_4)\mathbf{T}_3^{-1}(\theta_3).\end{aligned}\quad (11)$$

The axes of the third joint and the fourth joint intersect,

$$\begin{aligned}\mathbf{T}_L &= \mathbf{T}_1(\theta_1)\mathbf{T}_2(\theta_2)\mathbf{T}_3(\theta_3)\text{Rot}(-\alpha_3, x_3) \\ \mathbf{T}_R &= \mathbf{T}_6^{-1}(\theta_6)\mathbf{T}_5^{-1}(\theta_5)\mathbf{T}_4^{-1}(\theta_4).\end{aligned}\quad (12)$$

The axes of the fourth joint and the fifth joint intersect,

$$\begin{aligned}\mathbf{T}_L &= \mathbf{T}_2(\theta_2)\mathbf{T}_3(\theta_3)\mathbf{T}_4(\theta_4)\text{Rot}(-\alpha_4, x_4) \\ \mathbf{T}_R &= \mathbf{T}_1^{-1}(\theta_1)\mathbf{T}_6^{-1}(\theta_6)\mathbf{T}_5^{-1}(\theta_5).\end{aligned}\quad (13)$$

The axes of the fifth joint and the sixth joint intersect,

$$\begin{aligned}\mathbf{T}_L &= \mathbf{T}_3(\theta_3)\mathbf{T}_4(\theta_4)\mathbf{T}_5(\theta_5)\text{Rot}(-\alpha_5, x_5) \\ \mathbf{T}_R &= \mathbf{T}_2^{-1}(\theta_2)\mathbf{T}_1^{-1}(\theta_1)\mathbf{T}_6^{-1}(\theta_6).\end{aligned}\quad (14)$$

According to the characteristics of the joint coordinate system established by the standard DH method, the first homogeneous matrix of the sub-chain L expressions in Eqs. (10)–(14) can be considered the base coordinate system two sub-chains. $\mathbf{T}$ and $\mathbf{T}^{-1}$ can be thought of as a virtual link between the origin of the original manipulator's base and EE.

In Eqs. (10) and (11), it is necessary to invert the target pose matrix. That will increase the computational overhead of subsequent inverse kinematics problems. Therefore, the above deformation modes need to be selected reasonably. Under the condition that the six-DOF manipulator has multiple intersecting points, the deformation mode can be chosen according to the following principles:

Principle 1: Preferential is given to the deformation mode of Eqs. (12)–(14).

Principle 2: Preferential selection allows the sub-chain L to contain the deformation of parallel joints.

Principle 3: Preferential is given to deformation, allowing each sub-chain end position equations to contain only two joint variables.

These three principles are parallel.

3.4. Inverse kinematics solution process

The six-DOF manipulator proposed in this paper can deduce each joint variable's IK formula according to the following steps.

Step 1: Select the deformation mode of the formula according to principle 1–3.

Step 2: In Eq. (4)–(5), obtain the IK solutions of the other four joint variables except for θ_i and θ_{i+1} using the existing elimination method of the three-DOF manipulator.

Step 3: Solve θ_i by using Eq. (6) or (8), and solve θ_{i+1} by using Eq. (7) or (9) based on the results of step 2.

4. Example verification

Three examples are developed in this section to show more details of the proposed method and its advantages. The trigonometric equations used in this section's derivation can be referred to in the appendix in the literature [55].

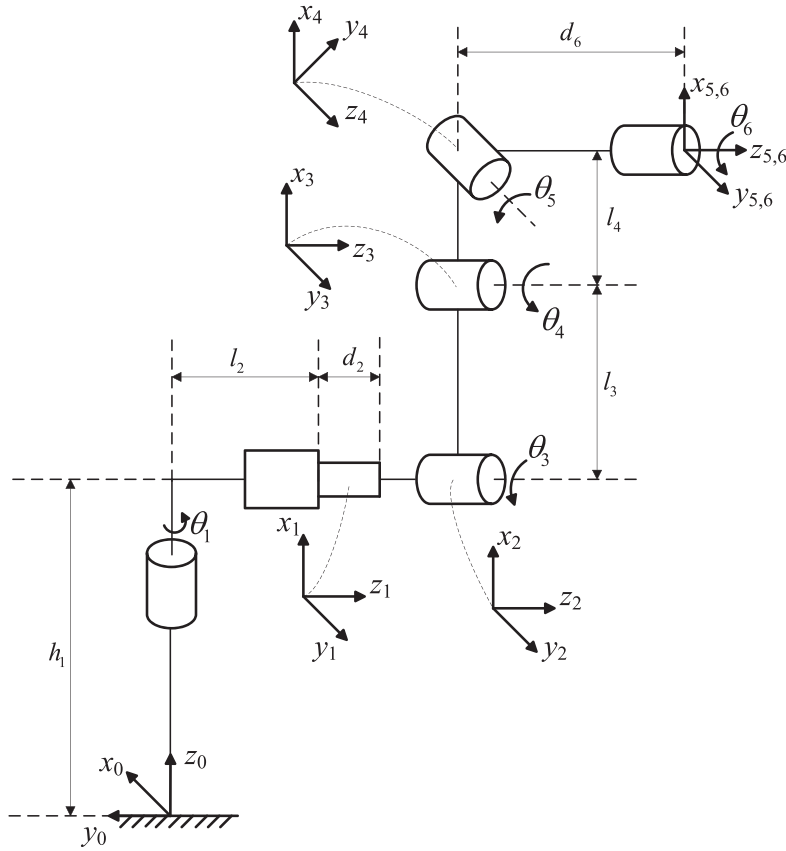


Fig. 4. Configuration of RC manipulator.

Table 1
DH parameters of RC manipulator.

i	θ_i (°)	d_i (mm)	a_i (mm)	α_i (°)
1	θ_1	h_1	0	90
2	90	d_2	0	0
3	θ_3	l_2	l_3	0
4	θ_4	0	l_4	-90
5	θ_5	0	0	90
6	θ_6	d_6	0	0

4.1. Derivation of inverse kinematics formula

4.1.1. RC manipulator

The rigid body and coordinate frame attachments of the RC manipulator [42] are illustrated in Fig. 4, and its D-H parameters are presented in Table 1.

The symbolic variables in Table 1 are substituted into Eq. (2) to obtain the forward kinematics formula of the RC manipulator EE. The target value of EE is a symbolic function in the form of Eq. (3). According to the three principles in Section 3.3, the origin of the third joint and the fifth joint in Fig. 4 are selected as the cut points, and the corresponding deformation is Eq. (14). Then, according to the conditional Eqs. (4) and (5) of reconnecting the two sub-chains, four equations containing only θ_1 , d_2 , θ_3 , and θ_4 can be obtained.

According to conditional Eq. (4)

$$l_4 c_{34} + l_3 c_3 = p_z - h_1 - a_2 d_6, \quad (15)$$

$$l_4 s_{34} + l_3 s_3 = (a_y d_6 - p_y) s_1 - (p_x - a_x d_6) c_1, \quad (16)$$

$$l_2 = (p_x - a_x d_6) s_1 - (p_y - a_y d_6) c_1 - d_2. \quad (17)$$

Table 2
DH parameters of the industrial manipulator.

i	θ_i (°)	d_i (mm)	a_i (mm)	α_i (°)
1	θ_1	0	l_1	90
2	θ_2	0	l_2	0
3	θ_3	0	l_3	90
4	θ_4	l_4	0	β
5	θ_5	l_5	0	$-\beta$
6	θ_6	l_6	0	0

According to conditional Eq. (5)

$$a_z s_{34} + c_{34}(a_x c_1 + a_y s_1) = 0, \quad (18)$$

from now on, s_i and c_i denote $\sin\theta_i$ and $\cos\theta_i$, s_{ij} and c_{ij} are denoted as $\sin(\theta_i + \theta_j)$ and $\cos(\theta_i + \theta_j)$, respectively.

Observe that Eqs. (15), (16), and (18) are independent of d_2 , but it is challenging to get analytic solutions. However, a nonlinear equation containing only θ_1 can be obtained by the following elimination.

First of all, put c_{34} and s_{34} on the right-hand side of Eqs. (15) and (16), respectively, and then calculate the sum of squares to get $(\theta_3 + \theta_4)$ represented by θ_1 .

$$\theta_{34} = \text{atan2}\left(k\sqrt{p_{Rx}^2 + p_{Ry}^2 - \left(\frac{l_4^2 + p_{Rx}^2 + p_{Ry}^2 - l_3^2}{2l_4}\right)^2}, \frac{l_4^2 + p_{Rx}^2 + p_{Ry}^2 - l_3^2}{2l_4}\right) + \text{atan2}(p_{Ry}, p_{Rx}), \quad (19)$$

where $k = \pm 1$, θ_{34} represents $(\theta_3 + \theta_4)$, p_{Rx} and p_{Ry} are represent the right-hand side of Eqs. (15) and (16), respectively.

A nonlinear equation containing only θ_1 can be obtained by substituting Eq. (19) into Eq. (18). For the convenience of description, Eqs. (19) and (18) are combined in order and denoted as function $f(\theta_1)$. We are then using a one-dimension search to find the zero points of $f(\theta_1)$ successively from zero in the interval $[0, 360^\circ]$, as shown in Fig. 5.

In the flowchart, the interval $[\theta_1^{k-1}, \theta_1^k]$ needs to be determined firstly by traversal so that $f(\theta_1^{k-1})f(\theta_1^k) < 0$, and then the bisection method is used in this interval. Since $k = \pm 1$, it is necessary to run the flowchart algorithm twice to find all θ_1 of the RC manipulator.

After all of θ_1 are found, the remaining joint variables can be obtained according to the following equations. The prismatic joint variable d_2 is obtained directly by Eq. (17)

$$d_2 = (p_x - a_x d_6)s_1 - (p_y - a_y d_6)c_1 - l_2. \quad (20)$$

The revolute joint variable θ_4 is obtained through the quadratic sum of Eqs. (15) and (16)

$$\theta_4 = \text{atan2}\left(k\sqrt{4l_3^2 l_4^2 - (p_{Rx}^2 + p_{Ry}^2 - l_3^2 - l_4^2)}, p_{Rx}^2 + p_{Ry}^2 - l_3^2 - l_4^2\right), \quad (21)$$

k is the same as that in Eq. (19).

Then

$$\theta_3 = \theta_{34} - \theta_4. \quad (22)$$

After that, the revolute joint variables θ_5 and θ_6 can be obtained, respectively, according to the conditional formulas (8) and (9).

$$\theta_5 = \text{atan2}(-1, 0) - \text{atan2}(a_y c_1 - a_x s_1, -a_z c_{34} + s_{34}(a_x c_1 + a_y s_1)), \quad (23)$$

$$\theta_6 = \text{atan2}(1, 0) - \text{atan2}(-c_{34}(a_x c_1 + a_y s_1) - o_z s_{34}, -n_z s_{34} - c_{34}(n_x c_1 + n_y s_1)), \quad (24)$$

4.1.2. Industrial manipulator with skew offset wrist

The rigid body and coordinate frame attachments of the industrial manipulator with skew offset wrist are illustrated in Fig. 6, and its D-H parameters are shown in Table 2.

According to the three principles in Section 3.3, the origin of the second joint and the fourth joint in Fig. 6 are selected as the cut points, and the corresponding deformation is Eq. (13). Then, according to the conditional Eqs. (4) and (5) of reconnecting the two sub-chains, four equations containing only θ_1 , θ_2 , θ_3 , and θ_6 can be obtained.

According to conditional Eq. (5)

$$s_{23}(-c_1 A_{23} - s_1 B_{23}) - c_{23} C_{23} = \cos \beta, \quad (25)$$

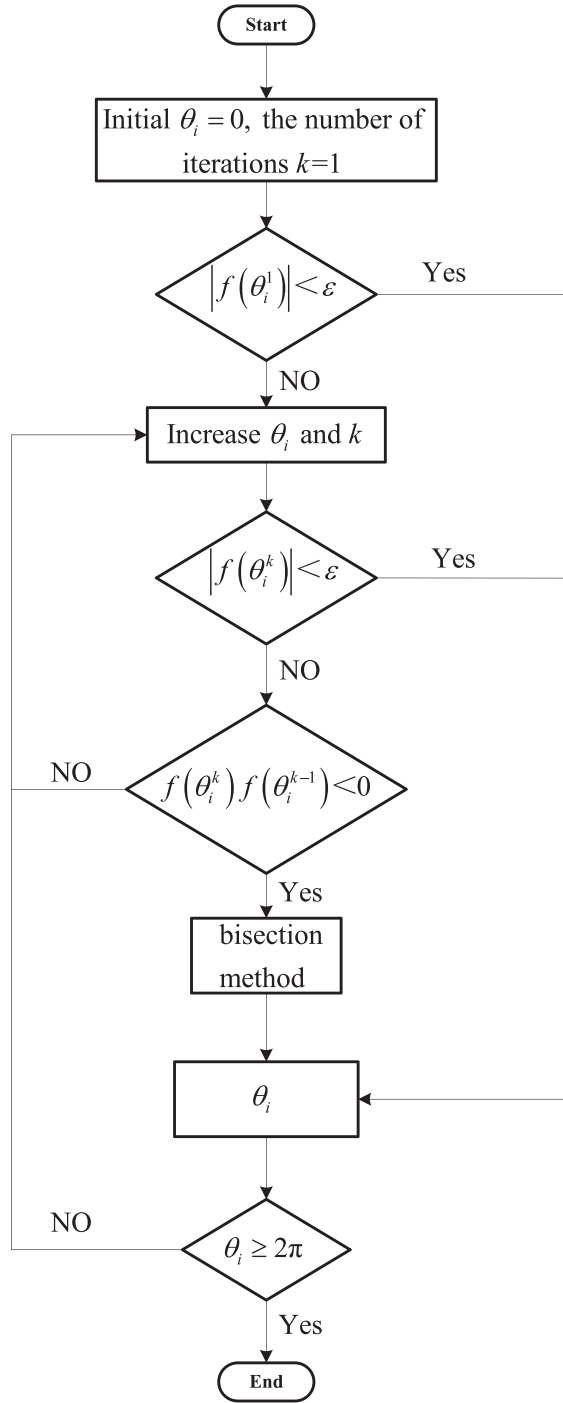


Fig. 5. The flowchart of finding θ_i .

where

$$A_{23} = -a_x c_\beta + o_x c_6 s_\beta + n_x s_\beta s_6, \quad (26)$$

$$B_{23} = -a_y c_\beta + o_y c_6 s_\beta + n_y s_\beta s_6, \quad (27)$$

$$C_{23} = a_z c_\beta - o_z c_6 s_\beta - n_z s_\beta s_6. \quad (28)$$

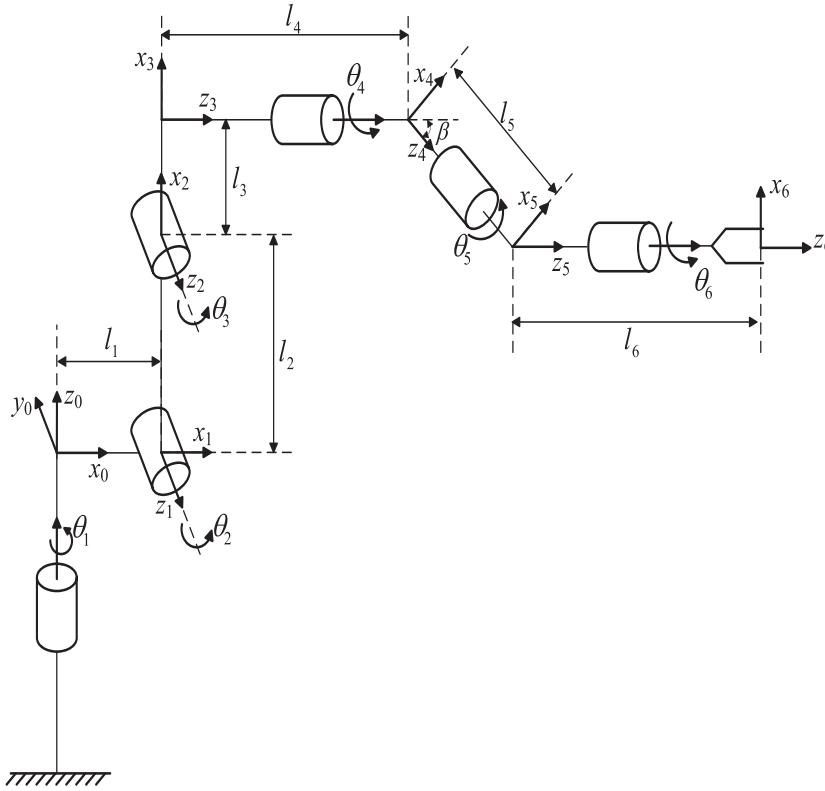


Fig. 6. Configuration of the industrial manipulator.

According to conditional Eq. (4)

$$l_3 c_{23} + l_4 s_{23} + l_2 c_2 = -l_1 + s_1(p_y - a_y l_6 + l_5 B_{23}) + c_1(p_x - a_x l_6 + l_5 A_{23}), \quad (29)$$

$$l_3 s_{23} - l_4 c_{23} + l_2 s_2 = pz - a_z l_6 - l_5 C_{23}, \quad (30)$$

$$0 = s_1(p_x - a_x l_6 + l_5 A_{23}) - c_1(p_y - a_y l_6 + l_5 B_{23}). \quad (31)$$

It is observed that in Eq. (31), θ_1 can be directly expressed by θ_6

$$\theta_1 = \text{atan2}(0, k_1) + \text{atan2}(p_y - a_y l_6 + l_5 B_{23}, p_x - a_x l_6 + l_5 A_{23}), \quad (32)$$

where $k_1 = \pm 1$.

Analogous to the previous section, obtained $(\theta_2 + \theta_3)$

$$\theta_{23} = \text{atan2}\left(k_2 \sqrt{A^2 + B^2 - C^2}, C\right) + \text{atan2}(B, A), \quad (33)$$

where $k_2 = \pm 1$, and

$$A = 2l_3 p_{Rx} - 2l_4 p_{Ry}, \quad (34)$$

$$B = 2l_3 p_{Ry} + 2l_4 p_{Rx}, \quad (35)$$

$$C = l_3^2 + l_4^2 - l_2^2 + p_{Rx}^2 + p_{Ry}^2. \quad (36)$$

wherein θ_{23} represents $(\theta_2 + \theta_3)$, p_{Rx} and p_{Ry} are represent the right-hand side of Eq. (29) and (30), respectively.

Similarly, Eqs. (32), (33), and (25) are combined in order and denoted as function $f(\theta_6)$. All solutions to θ_6 can be found by compiling the algorithm according to the flowchart in Fig. 5. Since there are four combinations of k_1 and k_2 , the flowchart needs to be compiled four times.

Table 3
All IK solutions of RC manipulator.

No.	θ_1 (°)	d_2 (mm)	θ_3 (°)	θ_4 (°)	θ_5 (°)	θ_6 (°)
1	27.9193	-66.7795	52.6568	-16.4841	-223.5886	262.2970
2	-100.0134	15.0109	54.9808	-24.9973	39.9836	-0.0012
3	79.9866	-67.0109	-54.9808	24.9973	-219.9836	179.9988
4	-152.0807	14.7795	-52.6568	16.4841	43.5886	82.2970

After all of θ_6 are found, obtained

$$\theta_3 = \text{atan2}\left(k_2\sqrt{l_3^2 + l_4^2 - C_3^2}, C_3\right) + \text{atan2}(l_4, l_3), \quad (37)$$

$$C_3 = \frac{p_{Kx}^2 + p_{Ky}^2 - l_2^2 - l_3^2 - l_4^2}{2l_2}, \quad (38)$$

k_2 is the same as that in Eq. (33).

Then got

$$\theta_2 = \theta_{23} - \theta_3. \quad (39)$$

Finally, the revolute joint variables θ_4 and θ_5 can be obtained according to the conditional formulas (8) and (9).

$$\theta_4 = \text{atan2}(0, s_\beta) + \text{atan2}(s_{23}C_{23} - (c_1A_{23} + s_1B_{23})C_{23}, s_1A_{23} - c_1B_{23}), \quad (40)$$

$$\theta_5 = \text{atan2}(0, s_\beta) + \text{atan2}(B_5, A_5). \quad (41)$$

where

$$A_5 = s_{23}(a_x c_1 s_\beta + a_y s_\beta s_1 + (o_x c_1 c_6 + n_x c_1 s_6 + o_y c_6 s_1 + n_y s_1 s_6) c_\beta) - c_{23}(a_z s_\beta + o_z c_\beta c_6 + n_z c_\beta s_6) \quad (42)$$

$$B_5 = [(n_x c_6 - o_x s_6) c_1 + (n_y c_6 - o_y s_6) s_1] s_{23} - (n_z c_6 - o_z s_6) c_{23}. \quad (43)$$

4.1.3. Spherical wrist industrial manipulator

Let $\beta=90$, $d_5=0$, then the manipulator in the previous section become a spherical wrist industrial manipulator. At this point, θ_1 directly leads to an analytic expression with two solutions from Eq. (32). The equations for other joint variables also become analytic. θ_6 can be obtained from Eq. (25)

$$\theta_6 = \text{atan2}(k_3, 0) + \text{atan2}(n_z c_{23} - (n_y s_1 + n_x c_1) s_{23}, o_z c_{23} - (o_y s_1 + o_x c_1) s_{23}). \quad (44)$$

where $k_3=\pm 1$.

There are eight sets of analytic solutions, with the positive and negative signs of k_1 , k_2 , and k_3 . The solution sequence of each joint variable is as follows (32)→(33)→(44)→(37)→(39)→(40)→(41).

4.2. Numerical simulation

Numerical simulations were provided to verify the algorithm. Set the threshold $\varepsilon=10^{-10}$ and the step size $h=0.5^\circ$ of Fig. 5. The computer programs were developed on Intel i3-3240@ 3.40GHz, RAM 4GB@ 1600MHz, and ran 1000 times.

The link parameters and the EE pose of RC manipulator were set to be consistent with those in literature [42], $h_1=38$, $l_2=26$, $l_3=18$, $l_4=10$, $l_6=10$, $n_x=0.699$, $o_x=0.15$, $a_x=-0.698$, $n_y=0.265$, $o_y=0.852$, $a_y=0.449$, $n_z=0.663$, $o_z=-0.5$, $a_z=0.556$, $p_x=-43.934$, $p_y=31.059$, $p_z=62.551$. Its all IK solutions were listed in Table 3.

Since the target pose given in the literature [42] is only three floating-point numbers, the 2-norm is 0.014, which derives from the error norm between the pose corresponding to the IK solution and the target pose in Table 3. There are four set IK solutions total, where the second set is consistent with that in [42]. Besides, the average time for all IK solutions is about 0.27ms.

The industrial manipulator's link parameters with skew offset wrist were set to be consistent with those in literature [41], $l_1=270$, $l_2=1300$, $l_3=42.5$, $l_4=1300$, $l_5=108.9$, $l_6=82$, $\beta=70^\circ$. Given a set of joint values $\theta=\{\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6\}=\{10^\circ, 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ\}$ to get the EE pose. Its all IK solutions are listed in Table 4.

As expected, all the IK solutions in the table are consistent with the real number of solutions in [41]. The maximum 2-norm is 8.001×10^{-9} , which derives from the error norm between the pose corresponding to IK solutions and the target pose. At the same time, the average time for all IK solutions is about 0.65ms.

Table 4

All IK solutions of the industrial manipulator with offset wrist.

No.	θ_1 (°)	θ_2 (°)	θ_3 (°)	θ_4 (°)	θ_5 (°)	θ_6 (°)
1	9.51656177	-36.20185327	142.49231019	11.70547721	99.04679162	85.77350772
2	13.02910837	-36.09097780	146.79087391	-124.76372620	-101.20283024	315.42226156
3	10.00000000	20.00000000	30.00000000	40.00000000	50.00000000	60.00000000
4	13.04472568	19.77641023	36.74862285	-125.24556456	-52.86920298	267.36249784

4.3. Discussion

In the previous content, the method presented in this paper works well from the determination of closure equations, derivation of IK formulas, and finally, numerical calculation.

The transformation form of FKE is determined intuitively by using double-cut points. Six corresponding derivation formulas can be found by using the condition of re-connection of sub-chains after transformation. There is no need to look for complex geometric relations as in literature [30] or make two transformations of FKE as in literature [42]. Besides, the manipulator examples in literature [42,49] have only one intersection point where the last two joint axes intersect. Therefore, their FKE basic transformation form is included in our proposed method.

After determining the closure equations, we need at most four equations with no more than four joint variables to get a univariate nonlinear equation. That single joint variable analytically expresses all other joint variables. The univariate nonlinear equation is consistent with the solution hypothesis proposed in the literature [42,48,49]. However, in our method's derivation process, fewer equations are used, and the expression form is more concise. Literature [42] obtained the unitary 20-degree polynomial of RC manipulator concerning θ_1 . Still, the coefficient expression of this polynomial is very complicated. As stated in its introduction, "obtaining the roots of 16-degree polynomial is very tedious and very time consuming", not to mention the roots of 20-degree polynomial.

At the same time, the derivation in this paper has obvious geometric significance. Eqs. (19) and (33) can be thought of as the ends of two sub-chains reconnected first. If the square root of them has the solution, there will be an infinite number of such connections. Then, solving the univariate nonlinear equation is to search for the solutions satisfying the conditional Eq. (5) in these cases. If the square root of them has no resolution, the two sub-chains can not be reconnected. That is, there is no IK solution to the EE pose. There is no such geometric meaning in literature [42,48,49].

The one-dimensional search algorithm proposed in this paper can meet the requirements of real-time and high precision. Since the search is carried out in the real domain when Eqs. (19) and (33) have no solution, the algorithm will skip. Hence, there is no solution in a complex form. We used the results in Refs. [42] for the same one-dimensional ergodic search, which required a minimum of 0.8ms, not including excluding the extra solutions. Simultaneously, our algorithm only needs about 0.27ms to obtain all the IK solutions of its given EE pose. In a similar computer configuration, the average time is about 0.65ms of all IK solutions of skew offset wrist industrial manipulator obtained in this paper, compared with 6.993ms in literature [41]. The one-dimensional search proposed in this paper is not an optimal solution. If the step size is too large, the necessary IK solution will be missed. If the step size is too small, the search time will be increased. The time consumption of the algorithm is mainly derived from the nonlinear equation calculation, and the time consumption can be reduced by reducing the number of its calculations.

From the formula derived previously, it can be seen that the wrist joint variables of the manipulators are not affected by wrist singularity. That is very important for the motion control of the industrial manipulators with spherical wrist, while the wrist singularity problem was not solved well in [43,44,56]. Besides, the time consumption and accuracy of solving the spherical wrist manipulator using our method are not different from those in literature [56].

5. Conclusion

The improved disconnection and re-connection method proposed in this paper can efficiently and uniformly derive IK formulas of the general 6R manipulator with simple geometry. (i) The first cut point is selected at the intersection point of adjacent joint axes. The sub-chains attitude component is set to be collinear with the joint axis, minimizing the number of variables in the closure equations. (ii) The second cut point is chosen to obtain two three-DOF sub-chains, which can reduce formula derivation difficulty. (iii) The new condition for the sub-chains' re-connection is determined as: the sub-chains' endpoints position is coincident. The included angle of their attitude remains unchanged compared with that before disconnection. (iv) The equations used to solve the IK problem can be directly determined according to the re-connection condition. (v) The six joint variables of the manipulator can be solved with only six equations. (vi) A univariate nonlinear equation can be obtained for structures without analytic solutions, and a one-dimensional search can find all its solutions. (vii) The solutions of wrist joint variables are not affected by singularity.

In the future, these results will be combined with the motion control of the manipulator.

Declaration of Competing Interest

None.

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